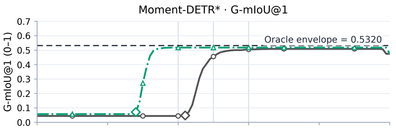
# 11 backbones · GMR baseline vs GEQ+aux

Fixed Val-selected checkpoints → Standard Test (1,036 queries)

Diamonds: Val ≥95% retention operating points (0.01 grid); ΔG1/G3 refer to these points.

Seed 2023 · 128 video tokens · GMR patience 20 / GEQ patience 10 · * Moment-DETR reused

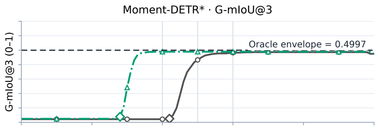
### Moment-DETR* · G-mIoU@1



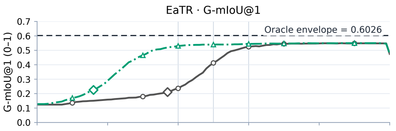

GMR baseline | mAP 5.49% | τ95 0.42 — GEQ + aux | mAP 7.47% | τ95 0.28

ΔmAP: +1.98 pp — ΔG1 at τ95: +2.33 pp — ΔG3 at τ95: +1.34 pp

### Moment-DETR* · G-mIoU@3



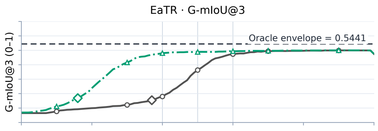

### EaTR · G-mIoU@1



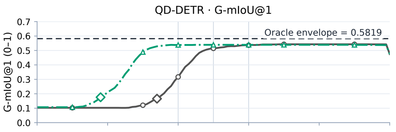

GMR baseline | mAP 16.37% | τ95 0.37 — GEQ + aux | mAP 17.16% | τ95 0.16

ΔmAP: +0.79 pp — ΔG1 at τ95: +1.46 pp — ΔG3 at τ95: +1.42 pp

### EaTR · G-mIoU@3



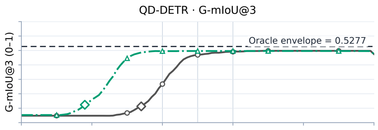

### QD-DETR · G-mIoU@1



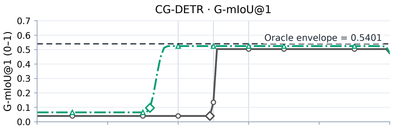

GMR baseline | mAP 11.56% | τ95 0.34 — GEQ + aux | mAP 13.05% | τ95 0.18

ΔmAP: +1.49 pp — ΔG1 at τ95: +0.99 pp — ΔG3 at τ95: +1.09 pp

### QD-DETR · G-mIoU@3



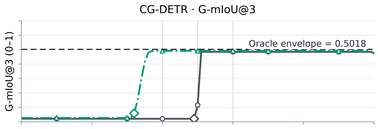

### CG-DETR · G-mIoU@1



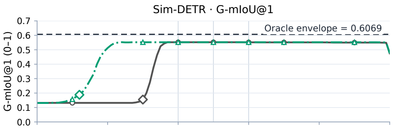

GMR baseline | mAP 6.37% | τ95 0.49 — GEQ + aux | mAP 6.91% | τ95 0.32

ΔmAP: +0.54 pp — ΔG1 at τ95: +5.60 pp — ΔG3 at τ95: +3.75 pp

### CG-DETR · G-mIoU@3



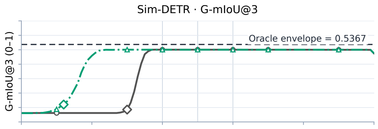

### Sim-DETR · G-mIoU@1



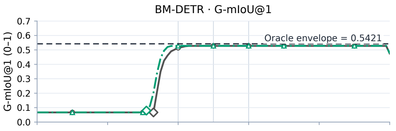

GMR baseline | mAP 20.29% | τ95 0.30 — GEQ + aux | mAP 21.00% | τ95 0.12

ΔmAP: +0.71 pp — ΔG1 at τ95: +3.32 pp — ΔG3 at τ95: +3.51 pp

### Sim-DETR · G-mIoU@3



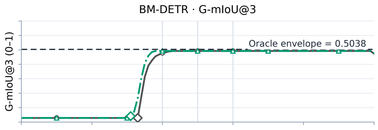

### BM-DETR · G-mIoU@1



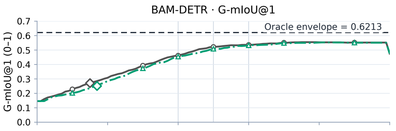

GMR baseline | mAP 4.11% | τ95 0.33 — GEQ + aux | mAP 4.20% | τ95 0.31

ΔmAP: +0.09 pp — ΔG1 at τ95: +1.26 pp — ΔG3 at τ95: +1.28 pp

### BM-DETR · G-mIoU@3



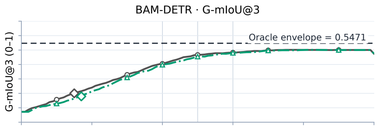

### BAM-DETR · G-mIoU@1



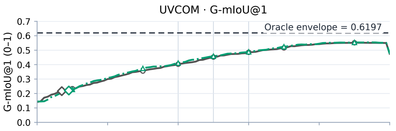

GMR baseline | mAP 22.97% | τ95 0.15 — GEQ + aux | mAP 23.64% | τ95 0.17

ΔmAP: +0.67 pp — ΔG1 at τ95: -1.99 pp — ΔG3 at τ95: -2.11 pp

### BAM-DETR · G-mIoU@3



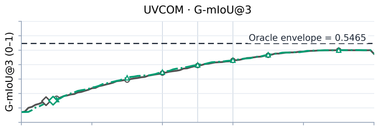

### UVCOM · G-mIoU@1



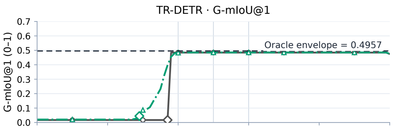

GMR baseline | mAP 22.21% | τ95 0.07 — GEQ + aux | mAP 21.66% | τ95 0.09

ΔmAP: -0.55 pp — ΔG1 at τ95: +0.44 pp — ΔG3 at τ95: +0.35 pp

### UVCOM · G-mIoU@3



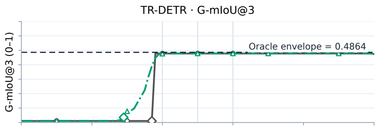

### TR-DETR · G-mIoU@1



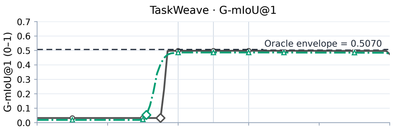

GMR baseline | mAP 2.58% | τ95 0.37 — GEQ + aux | mAP 2.81% | τ95 0.29

ΔmAP: +0.23 pp — ΔG1 at τ95: +2.72 pp — ΔG3 at τ95: +2.48 pp

### TR-DETR · G-mIoU@3



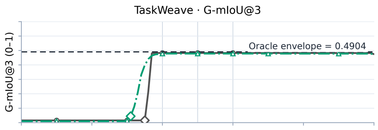

### TaskWeave · G-mIoU@1



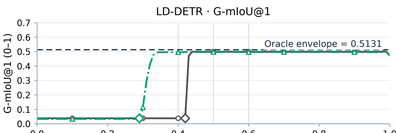

GMR baseline | mAP 3.79% | τ95 0.35 — GEQ + aux | mAP 3.40% | τ95 0.31

ΔmAP: -0.39 pp — ΔG1 at τ95: +2.10 pp — ΔG3 at τ95: +2.89 pp

### TaskWeave · G-mIoU@3



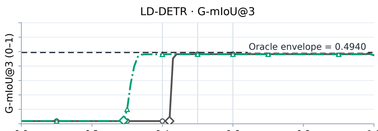

### LD-DETR · G-mIoU@1




GMR baseline | mAP 6.32% | τ95 0.42 — GEQ + aux | mAP 6.52% | τ95 0.29

ΔmAP: +0.20 pp — ΔG1 at τ95: -0.05 pp — ΔG3 at τ95: +0.40 pp

### LD-DETR · G-mIoU@3




Oracle envelope: max of the two methods with GT presence and each method's fixed candidates; analysis only.

Curves: 101 fixed thresholds, 0–1 score units. τ95 uses Standard Val only; the same threshold is used on both Tests.

mAP is ungated; Δ = GEQ+aux − matching GMR baseline (auxiliary-trained Stage 1). No Test tuning or model reselection.

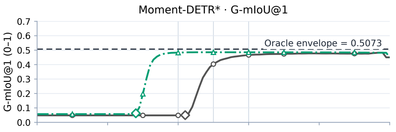
# 11 backbones · GMR baseline vs GEQ+aux

Fixed Val-selected checkpoints → Full-Decodable Test (2,970 / 2,986 queries)

Diamonds: Val ≥95% retention operating points (0.01 grid); ΔG1/G3 refer to these points.

Seed 2023 · 128 video tokens · GMR patience 20 / GEQ patience 10 · * Moment-DETR reused

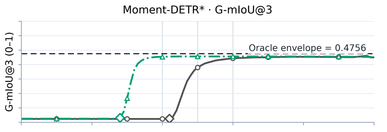
### Moment-DETR* · G-mIoU@1



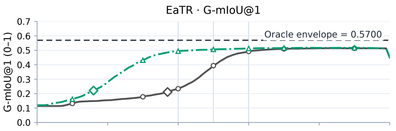

GMR baseline | mAP 5.33% | τ95 0.42 — GEQ + aux | mAP 6.44% | τ95 0.28

ΔmAP: +1.11 pp — ΔG1 at τ95: +1.29 pp — ΔG3 at τ95: +0.57 pp

### Moment-DETR* · G-mIoU@3



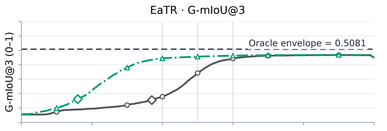

### EaTR · G-mIoU@1



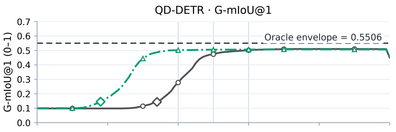

GMR baseline | mAP 13.77% | τ95 0.37 — GEQ + aux | mAP 14.68% | τ95 0.16

ΔmAP: +0.91 pp — ΔG1 at τ95: +0.99 pp — ΔG3 at τ95: +0.66 pp

### EaTR · G-mIoU@3



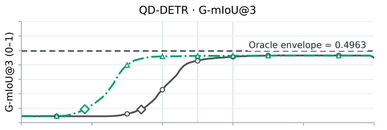

### QD-DETR · G-mIoU@1



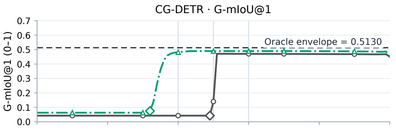

GMR baseline | mAP 10.50% | τ95 0.34 — GEQ + aux | mAP 11.22% | τ95 0.18

ΔmAP: +0.72 pp — ΔG1 at τ95: +0.17 pp — ΔG3 at τ95: +0.22 pp

### QD-DETR · G-mIoU@3



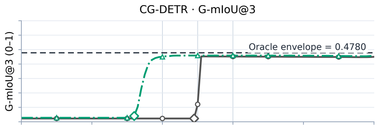

### CG-DETR · G-mIoU@1



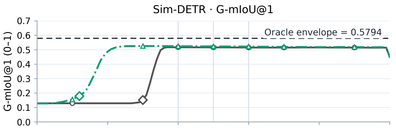

GMR baseline | mAP 5.82% | τ95 0.49 — GEQ + aux | mAP 6.19% | τ95 0.32

ΔmAP: +0.37 pp — ΔG1 at τ95: +3.21 pp — ΔG3 at τ95: +1.69 pp

### CG-DETR · G-mIoU@3



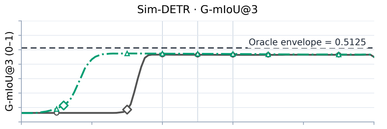

### Sim-DETR · G-mIoU@1



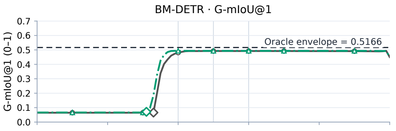

GMR baseline | mAP 19.60% | τ95 0.30 — GEQ + aux | mAP 19.46% | τ95 0.12

ΔmAP: -0.14 pp — ΔG1 at τ95: +2.70 pp — ΔG3 at τ95: +2.89 pp

### Sim-DETR · G-mIoU@3



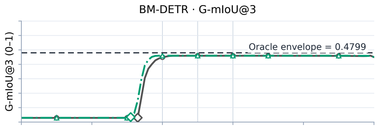

### BM-DETR · G-mIoU@1



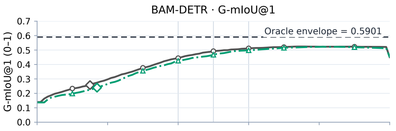

GMR baseline | mAP 3.87% | τ95 0.33 — GEQ + aux | mAP 3.81% | τ95 0.31

ΔmAP: -0.06 pp — ΔG1 at τ95: +0.53 pp — ΔG3 at τ95: +0.43 pp

### BM-DETR · G-mIoU@3



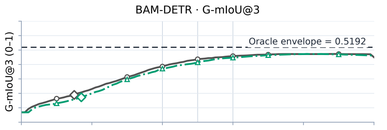

### BAM-DETR · G-mIoU@1



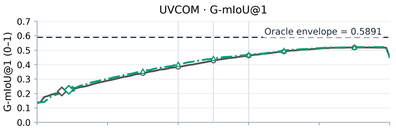

GMR baseline | mAP 20.33% | τ95 0.15 — GEQ + aux | mAP 20.59% | τ95 0.17

ΔmAP: +0.26 pp — ΔG1 at τ95: -1.77 pp — ΔG3 at τ95: -1.63 pp

### BAM-DETR · G-mIoU@3



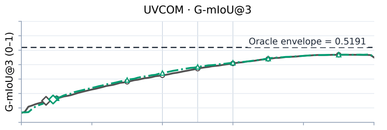

### UVCOM · G-mIoU@1



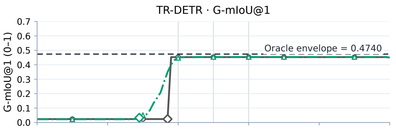

GMR baseline | mAP 18.46% | τ95 0.07 — GEQ + aux | mAP 19.21% | τ95 0.09

ΔmAP: +0.75 pp — ΔG1 at τ95: +1.16 pp — ΔG3 at τ95: +0.92 pp

### UVCOM · G-mIoU@3



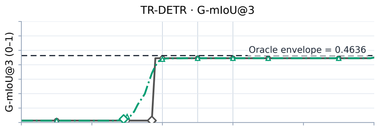

### TR-DETR · G-mIoU@1



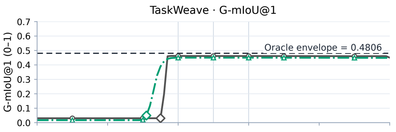

GMR baseline | mAP 2.85% | τ95 0.37 — GEQ + aux | mAP 2.84% | τ95 0.29

ΔmAP: -0.01 pp — ΔG1 at τ95: +0.92 pp — ΔG3 at τ95: +1.00 pp

### TR-DETR · G-mIoU@3



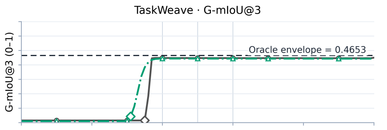

### TaskWeave · G-mIoU@1



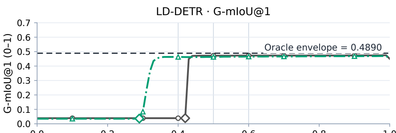

GMR baseline | mAP 3.70% | τ95 0.35 — GEQ + aux | mAP 3.31% | τ95 0.31

ΔmAP: -0.39 pp — ΔG1 at τ95: +1.89 pp — ΔG3 at τ95: +2.74 pp

### TaskWeave · G-mIoU@3



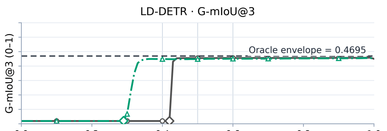

### LD-DETR · G-mIoU@1




GMR baseline | mAP 6.06% | τ95 0.42 — GEQ + aux | mAP 6.31% | τ95 0.29

ΔmAP: +0.25 pp — ΔG1 at τ95: -0.24 pp — ΔG3 at τ95: +0.13 pp

### LD-DETR · G-mIoU@3




Oracle envelope: max of the two methods with GT presence and each method's fixed candidates; analysis only.

Curves: 101 fixed thresholds, 0–1 score units. τ95 uses Standard Val only; the same threshold is used on both Tests.

mAP is ungated; Δ = GEQ+aux − matching GMR baseline (auxiliary-trained Stage 1). No Test tuning or model reselection.